```
praveen@PRAVEENKUMAR:~$ sudo apt install -y xz-utils gzip bzip2 tar zip unzip
[sudo: authenticate] Password:
xz-utils is already the newest version (5.8.3-1ubuntu0.1).
xz-utils set to manually installed.
gzip is already the newest version (1.14-1~exp2ubuntu1.1).
bzip2 is already the newest version (1.0.8-6ubuntu0.1).
tar is already the newest version (1.35+dfsg-4ubuntu0.4).
zip is already the newest version (3.0-15ubuntu3).
unzip is already the newest version (6.0-29ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 31
praveen@PRAVEENKUMAR:~$ mkdir -p ~/compression_practice/docs
praveen@PRAVEENKUMAR:~$ mkdir -p ~/compression_practice/site/css
praveen@PRAVEENKUMAR:~$ mkdir -p ~/compression_practice/site/js
praveen@PRAVEENKUMAR:~$ cd ~/compression_practice
praveen@PRAVEENKUMAR:~/compression_practice$ echo "This is a sample log file for compression testing" > big.log
praveen@PRAVEENKUMAR:~/compression_practice$ echo "Document 1 content" > docs/file1.txt
praveen@PRAVEENKUMAR:~/compression_practice$ echo "Document 2 content" > docs/file2.txt
praveen@PRAVEENKUMAR:~/compression_practice$ echo "Application log data" > docs/app.log
praveen@PRAVEENKUMAR:~/compression_practice$ echo "<html><body>Welcome to my website</body></html>" > site/index.html
praveen@PRAVEENKUMAR:~/compression_practice$ echo "body { margin: 0; }" > site/css/style.css
praveen@PRAVEENKUMAR:~/compression_practice$ echo "console.log('Hello World');" > site/js/app.js
praveen@PRAVEENKUMAR:~/compression_practice$ xz big.log
praveen@PRAVEENKUMAR:~/compression_practice$ ls -lh big.log.xz
-rw-r--r-- 1 praveen praveen 116 Nov  5 10:21 big.log.xz
praveen@PRAVEENKUMAR:~/compression_practice$ unxz big.log.xz
praveen@PRAVEENKUMAR:~/compression_practice$ gzip -k big.log
praveen@PRAVEENKUMAR:~/compression_practice$ zcat big.log.gz | head
This is a sample log file for compression testing
praveen@PRAVEENKUMAR:~/compression_practice$ zless big.log.gz
praveen@PRAVEENKUMAR:~/compression_practice$ tar -cjf b.tar.bz2 docs/
praveen@PRAVEENKUMAR:~/compression_practice$ ls -lh b.tar.bz2
-rw-r--r-- 1 praveen praveen 249 Nov  5 10:22 b.tar.bz2
praveen@PRAVEENKUMAR:~/compression_practice$ tar -czf b.tgz --exclude='*.log' docs/
praveen@PRAVEENKUMAR:~/compression_practice$ tar -tzf b.tgz
docs/
docs/file2.txt
docs/file1.txt
praveen@PRAVEENKUMAR:~/compression_practice$ tar -tzf b.tgz
docs/
docs/file2.txt
docs/file1.txt
praveen@PRAVEENKUMAR:~/compression_practice$ tar -xzf b.tgz -C /tmp
praveen@PRAVEENKUMAR:~/compression_practice$ ls -R /tmp/docs
/tmp/docs:
file1.txt  file2.txt
praveen@PRAVEENKUMAR:~/compression_practice$ zip -r site.zip site/
```

```
file1.txt  file2.txt
praveen@PRAVEENKUMAR:~/compression_practice$ zip -r site.zip site/
  adding: site/ (stored 0%)
  adding: site/js/ (stored 0%)
  adding: site/js/app.js (stored 0%)
  adding: site/index.html (deflated 8%)
  adding: site/css/ (stored 0%)
  adding: site/css/style.css (stored 0%)
praveen@PRAVEENKUMAR:~/compression_practice$ ls -lh site.zip
-rw-r--r-- 1 praveen praveen 1020 Nov  5 10:23 site.zip
praveen@PRAVEENKUMAR:~/compression_practice$ unzip -l site.zip
Archive:  site.zip
  Length      Date    Time    Name
---------  ---------- -----   ----
        0  2026-11-05 10:21   site/
        0  2026-11-05 10:21   site/js/
       28  2026-11-05 10:21   site/js/app.js
       48  2026-11-05 10:21   site/index.html
        0  2026-11-05 10:21   site/css/
       20  2026-11-05 10:21   site/css/style.css
---------                     -------
       96                     6 files
praveen@PRAVEENKUMAR:~/compression_practice$ unzip site.zip -d /tmp/site
Archive:  site.zip
   creating: /tmp/site/site/
   creating: /tmp/site/site/js/
 extracting: /tmp/site/site/js/app.js
  inflating: /tmp/site/site/index.html
   creating: /tmp/site/site/css/
 extracting: /tmp/site/site/css/style.css
praveen@PRAVEENKUMAR:~/compression_practice$ ls -R /tmp/site
/tmp/site:
site

/tmp/site/site:
css  index.html  js

/tmp/site/site/css:
style.css

/tmp/site/site/js:
app.js
praveen@PRAVEENKUMAR:~/compression_practice$
```